

Regularized Choi-Covariance Contractions for the Class-A Adaptive Fermionic Circuit

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x We formalize the regulated Gaussian contraction calculation underlying a cumulative Choi-covariance representation of realized trajectories in the class-A adaptive fermionic circuit. A local rank-one Wannier-mode operation is represented by a two-leg covariance tensor $K(\eta_1, \eta_2) = \begin{pmatrix} \eta_1 P & \mathbb{I} - P \\ \mathbb{I} - P & \eta_2 P \end{pmatrix}$, where $P = |\chi\rangle\langle\chi|$ and $\eta_1, \eta_2 \in \{\pm 1\}$ select the four measurement or correction branches. Direct composition with an accumulated Choi covariance is apparently obstructed because $K_{LL} = \eta_1 P$ is singular on the orthogonal complement of the measured mode. Introducing $K_{LL}^{(\varepsilon)} = \eta_1 P + \varepsilon(\mathbb{I} - P)$, performing the Schur-complement contraction, and subsequently taking $\varepsilon \rightarrow 0$ produces a finite update whenever $\text{Tr}(P\Sigma_{RR}) - \eta_1 \neq 0$. We give the blockwise calculation, correct a dimensional ambiguity in the lower-right block of the original handwritten derivation, and state the result entirely in basis-independent projector notation. This cumulative operator construction is distinct from the support-reduced state-covariance dynamics implemented by the repository’s canonical Markov-circuit engine.

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I. CIRCUIT-OPERATOR VIEWPOINT

Consider one realized sequence of local measurement and feedback events. Its unnormalized Gaussian circuit operator is denoted by $\hat{C}_t$, so that the normalized physical state at time step t is

$$|\psi_t\rangle = \frac{\hat{C}_t|\psi_0\rangle}{\|\hat{C}_t|\psi_0\rangle\|}, \quad \hat{C}_{t+1} = \hat{K}_{t+1}\hat{C}_t. \quad (1)$$

The object considered below is a two-leg Gaussian Choi covariance for $\hat{C}_t$,

$$\Sigma_t = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}, \quad \Sigma_{RL} = \Sigma_{LR}^\dagger. \quad (2)$$

The L and R labels refer to the input and output legs of the cumulative operator tensor, rather than to spatial regions.

This viewpoint must be distinguished from the state-covariance formulation used for production dynamics in this repository. There the dynamical variable is the top-layer covariance

$$G_{ij} = 2\langle c_i^\dagger c_j \rangle - \delta_{ij}, \quad (3)$$

which is propagated directly by `classA_U1FGTN.run_markov_circuit(...)` or its GPU analogue. The present calculation instead asks how a full realized operator $\hat{C}_t$ would be updated and retained as a Choi tensor.

II. LOCAL ADAPTIVE BRANCHES

Let $|\chi\rangle$ be a normalized single-particle Wannier mode and let

$$P = |\chi\rangle\langle\chi|, \quad Q = \mathbb{I} - P. \quad (4)$$

Thus P is rank one and Q projects onto all modes orthogonal to the locally interrogated channel. The four local Gaussian branches written in the handwritten calculation can be assembled into

$$K(\eta_1, \eta_2) = \begin{pmatrix} K_{LL} & K_{LR} \\ K_{RL} & K_{RR} \end{pmatrix} = \begin{pmatrix} \eta_1 P & Q \\ Q & \eta_2 P \end{pmatrix}, \quad \eta_1, \eta_2 \in \{+1, -1\}. \quad (5)$$

In the sign convention of the handwritten notes, the four branch assignments encoded by Eq. (5) are listed in Table I.

In the present implementation of the circuit, each site supplies the Observable-Wannier channels χ_{A+} , χ_{A-} , χ_{B+} , and χ_{B-} . The desired state empties the upper-band modes $A+$ and $B+$, while filling the lower-band modes $A-$ and $B-$. Consequently the table is the operator-level counterpart of the measurement and feedback logic implemented for `Ap`, `Am`, `Bp`, and `Bm`.

III. TWO-LEG COMPOSITION FORMULA

Contract the right leg of the accumulated tensor Σ with the left leg of a new local tensor K . The standard Gaussian two-leg contraction gives

$$\begin{aligned} \Sigma' &= K \circ \Sigma \\ &= \begin{pmatrix} \Sigma_{LL} & 0 \\ 0 & \eta_2 P \end{pmatrix} - \begin{pmatrix} \Sigma_{LR} & 0 \\ 0 & Q \end{pmatrix} \begin{pmatrix} \Sigma_{RR} & -\mathbb{I} \\ -\mathbb{I} & \eta_1 P \end{pmatrix}^{-1} \begin{pmatrix} \Sigma_{RL} & 0 \\ 0 & Q \end{pmatrix}. \end{aligned} \quad (6)$$

TABLE I. Rank-one local operations encoded by Eq. (5). The $+$ and $-$ labels on the mode indicate upper- and lower-band Observable-Wannier channels, not the signs $\eta_{1,2}$.

Operation	Physical role	(η_1, η_2)
$\hat{N}_-$	lower-band mode already occupied	$(-1, +1)$
$\hat{\chi}_-^\dagger (1 - \hat{N}_-)$	fill a lower-band vacancy	$(+1, +1)$
$1 - \hat{N}_+$	upper-band mode already empty	$(+1, -1)$
$\hat{\chi}_+ \hat{N}_+$	remove an upper-band particle	$(-1, -1)$

The expression is formally singular because the internal lower-right block $\eta_1 P$ has rank one. Inverting that block would require an inverse of P on $\text{Ran } Q$, where it vanishes identically.

To define the contraction as a limit, replace the singular block by

$$A_\varepsilon = \eta_1 P + \varepsilon Q, \quad A_\varepsilon^{-1} = \eta_1 P + \frac{1}{\varepsilon} Q, \quad \varepsilon \neq 0. \quad (7)$$

Let

$$R \equiv \Sigma_{RR}, \quad \mathcal{K}_\varepsilon = \begin{pmatrix} R & -\mathbb{I} \\ -\mathbb{I} & A_\varepsilon \end{pmatrix}. \quad (8)$$

For fixed nonzero ε , block inversion through the Schur complement

$$S_\varepsilon = R - A_\varepsilon^{-1} \quad (9)$$

yields

$$\mathcal{K}_\varepsilon^{-1} = \begin{pmatrix} S_\varepsilon^{-1} & S_\varepsilon^{-1} A_\varepsilon^{-1} \\ A_\varepsilon^{-1} S_\varepsilon^{-1} & A_\varepsilon^{-1} + A_\varepsilon^{-1} S_\varepsilon^{-1} A_\varepsilon^{-1} \end{pmatrix}. \quad (10)$$

Substitution into Eq. (6) reduces the problem to four regulated limits.

IV. CALCULATION IN A PROJECTOR-ADAPTED BASIS

Choose a temporary computational basis adapted to

$$\mathcal{H} = \text{Ran } P \oplus \text{Ran } Q, \quad \dim \text{Ran } P = 1. \quad (11)$$

In this basis,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{I}_Q \end{pmatrix}, \quad R = \begin{pmatrix} a & b^\dagger \\ b & C \end{pmatrix}, \quad (12)$$

where

$$a \in \mathbb{C}, \quad b \in \mathbb{C}^{(N-1) \times 1}, \quad C \in \mathbb{C}^{(N-1) \times (N-1)}. \quad (13)$$

For Hermitian R , a is real and $C = C^\dagger$. Define

$$d = a - \eta_1. \quad (14)$$

The following discussion assumes $d \neq 0$.

Using Eq. (7),

$$S_\varepsilon = \begin{pmatrix} d & b^\dagger \\ b & C - \varepsilon^{-1} \mathbb{I}_Q \end{pmatrix}. \quad (15)$$

Since

$$(C - \varepsilon^{-1} \mathbb{I}_Q)^{-1} = -\varepsilon (\mathbb{I}_Q - \varepsilon C)^{-1} = -\varepsilon \mathbb{I}_Q - \varepsilon^2 C + O(\varepsilon^3), \quad (16)$$

the block inverse of S_ε gives the finite combinations needed by the Choi contraction:

$$S_\varepsilon^{-1} \longrightarrow \begin{pmatrix} d^{-1} & 0 \\ 0 & 0 \end{pmatrix}, \quad (17)$$

$$S_\varepsilon^{-1} A_\varepsilon^{-1} Q \longrightarrow \begin{pmatrix} 0 & d^{-1} b^\dagger \\ 0 & -\mathbb{I}_Q \end{pmatrix}, \quad (18)$$

$$Q A_\varepsilon^{-1} S_\varepsilon^{-1} \longrightarrow \begin{pmatrix} 0 & 0 \\ d^{-1} b & -\mathbb{I}_Q \end{pmatrix}, \quad (19)$$

$$Q (A_\varepsilon^{-1} A_\varepsilon^{-1} S_\varepsilon^{-1} A_\varepsilon^{-1}) Q \longrightarrow \begin{pmatrix} 0 & 0 \\ 0 & d^{-1} b b^\dagger - C \end{pmatrix}. \quad (20)$$

Although the inverse A_ϵ^{-1} diverges on $\text{Ran } Q$, its divergent contribution cancels in the combinations selected by the external Q factors of the local tensor.

A. Block Updates in the Adapted Basis

Equations (17)–(20) immediately imply

$$\Sigma'_{LL} = \Sigma_{LL} - \Sigma_{LR} \begin{pmatrix} d^{-1} & 0 \\ 0 & 0 \end{pmatrix} \Sigma_{RL}, \quad (21)$$

$$\Sigma'_{LR} = \Sigma_{LR} \begin{pmatrix} 0 & -d^{-1}b^\dagger \\ 0 & \mathbb{I}_Q \end{pmatrix}, \quad (22)$$

$$\Sigma'_{RL} = \begin{pmatrix} 0 & 0 \\ -d^{-1}b & \mathbb{I}_Q \end{pmatrix} \Sigma_{RL}, \quad (23)$$

$$\Sigma'_{RR} = \boxed{\begin{pmatrix} \eta_2 & 0_{1 \times (N-1)} \\ 0_{(N-1) \times 1} & C - \frac{bb^\dagger}{a - \eta_1} \end{pmatrix}}. \quad (24)$$

V. DIMENSIONAL CORRECTION TO THE LOWER-RIGHT BLOCK

The handwritten calculation identifies the correct reduced Q -sector Schur complement,

$$C - \frac{bb^\dagger}{a - \eta_1}. \quad (25)$$

There is, however, an important dimensional qualification. The matrices C and $bb^\dagger$ in Eq. (25) act on $\text{Ran } Q$ and hence have dimensions $(N-1) \times (N-1)$. By contrast, P , Q , and Σ'_{RR} act on the full N -dimensional single-particle space. Thus an expression that adds $\eta_2 P$ directly to Eq. (25), or multiplies Eq. (25) by the full-space projector Q without declaring an embedding, is dimensionally incomplete.

The corrected statement is Eq. (24): the scalar η_2 occupies the P -sector, while the reduced Schur complement is embedded as the Q -sector block. In physical terms, the selected mode is reset to the covariance prescribed by η_2 , and the modes not selected by P inherit the conditional Schur-complement covariance induced by their previous correlations with that mode.

VI. BASIS-INDEPENDENT UPDATE

The adapted basis serves only to calculate the limit. Its final result can be written entirely using P , Q , and the original Choi blocks. First, the reduced quantities appearing above obey

$$a = \text{Tr}(PR) = \langle \chi | R | \chi \rangle, \quad (26)$$

$$b \longleftrightarrow QR|\chi\rangle, \quad (27)$$

$$bb^\dagger \longleftrightarrow QRPRQ, \quad (28)$$

$$C \longleftrightarrow QRQ|_{\text{Ran } Q}. \quad (29)$$

The arrows in Eqs. (27)–(29) indicate that the left-hand objects are matrices in a basis of $\text{Ran } Q$, whereas the right-hand expressions are their full-space operator embeddings.

Proposition 1 (Regulated rank-one Choi contraction). *Let $P = |\chi\rangle\langle\chi|$ be rank one, $Q = \mathbb{I} - P$, and*

$$K(\eta_1, \eta_2) = \begin{pmatrix} \eta_1 P & Q \\ Q & \eta_2 P \end{pmatrix}.$$

Let Σ be as in Eq. (2), write $R = \Sigma_{RR}$, and suppose

$$d = \text{Tr}(PR) - \eta_1 = \langle \chi | R | \chi \rangle - \eta_1 \neq 0. \quad (30)$$

Then the $\varepsilon \rightarrow 0$ limit of the contraction obtained from $K_{LL}^{(\varepsilon)} = \eta_1 P + \varepsilon Q$ exists and is

$$\Sigma' = \begin{pmatrix} \Sigma_{LL} - \frac{\Sigma_{LR}P\Sigma_{RL}}{d} & \Sigma_{LR}Q - \frac{\Sigma_{LR}PRQ}{d} \\ Q\Sigma_{RL} - \frac{QRP\Sigma_{RL}}{d} & \eta_2 P + QRQ - \frac{QRPQ}{d} \end{pmatrix}. \quad (31)$$

For clarity, the four operator-valued blocks of Eq. (31) are

$$\Sigma'_{LL} = \Sigma_{LL} - \frac{\Sigma_{LR}P\Sigma_{RL}}{\text{Tr}(P\Sigma_{RR}) - \eta_1}, \quad (32)$$

$$\Sigma'_{LR} = \Sigma_{LR}Q - \frac{\Sigma_{LR}P\Sigma_{RR}Q}{\text{Tr}(P\Sigma_{RR}) - \eta_1}, \quad (33)$$

$$\Sigma'_{RL} = Q\Sigma_{RL} - \frac{Q\Sigma_{RR}P\Sigma_{RL}}{\text{Tr}(P\Sigma_{RR}) - \eta_1}, \quad (34)$$

$$\Sigma'_{RR} = \eta_2 P + Q\Sigma_{RR}Q - \frac{Q\Sigma_{RR}P\Sigma_{RR}Q}{\text{Tr}(P\Sigma_{RR}) - \eta_1}. \quad (35)$$

Every term in Eqs. (32)–(35) is an operator on the full N -dimensional single-particle space. No dimension-dependent insertion maps or implied reduced-space products remain.

A. Sherman–Morrison Form of the RR Block

The Q -sector of Σ'_{RR} admits a compact inverse representation. Whenever $R = \Sigma_{RR}$ is invertible, the Sherman–Morrison formula applied to the rank-one perturbation $-\eta_1 P$ of R^{-1} gives

$$(R^{-1} - \eta_1 P)^{-1} = R + \frac{\eta_1 RPR}{1 - \eta_1 \langle \chi | R | \chi \rangle}. \quad (36)$$

Because $\langle \chi | R | \chi \rangle = d + \eta_1$ and $\eta_1^2 = 1$, the denominator collapses:

$$1 - \eta_1(d + \eta_1) = 1 - \eta_1 d - \eta_1^2 = -\eta_1 d, \quad (37)$$

so Eq. (36) becomes

$$(R^{-1} - \eta_1 P)^{-1} = R - \frac{RPR}{d}. \quad (38)$$

Projecting onto $\text{Ran } Q$ on both sides,

$$Q(R^{-1} - \eta_1 P)^{-1}Q = QRQ - \frac{QRPQ}{d}, \quad (39)$$

which is exactly the Q -sector of Σ'_{RR} in Eq. (35). The full updated right-right block therefore reads

$$\Sigma'_{RR} = \eta_2 P + Q(\Sigma_{RR}^{-1} - \eta_1 P)^{-1}Q. \quad (40)$$

Equation (40) makes the singular-branch condition transparent: the Sherman–Morrison denominator for inverting $\Sigma_{RR}^{-1} - \eta_1 P$ is $-\eta_1 d$, which vanishes if and only if $d = 0$. The regulated contraction is therefore well-defined on precisely those branches for which the inverse in Eq. (40) exists.

B. Regularized Resolvent Form of All Blocks

The inverse form in Eq. (40) can be written without first forming R^{-1} . Define the one-sided resolvents

$$A_R = \mathbb{I} - \eta_1 RP, \quad A_L = \mathbb{I} - \eta_1 PR. \quad (41)$$

If R is invertible, then

$$R^{-1} - \eta_1 P = R^{-1} A_R = A_L R^{-1}, \quad (42)$$

and hence

$$(R^{-1} - \eta_1 P)^{-1} = A_R^{-1} R = R A_L^{-1}. \quad (43)$$

The final resolvent formulas below only require the corresponding one-sided resolvent to be invertible; no explicit inverse of R need be formed in their numerical implementation. The Sherman–Morrison identity for $A_R = \mathbb{I} - \eta_1 RP$ gives

$$A_R^{-1} = \mathbb{I} - \frac{RP}{d}, \quad A_L^{-1} = \mathbb{I} - \frac{PR}{d}, \quad (44)$$

where $d = \text{Tr}(PR) - \eta_1$. Substitution into Eqs. (32)–(35) yields the equivalent resolvent representation

$$\boxed{\Sigma'_{LL} = \Sigma_{LL} + \eta_1 \Sigma_{LR} P A_R^{-1} \Sigma_{RL}} = \boxed{\Sigma_{LL} + \eta_1 \Sigma_{LR} A_L^{-1} P \Sigma_{RL}}, \quad (45)$$

$$\boxed{\Sigma'_{LR} = \Sigma_{LR} Q + \eta_1 \Sigma_{LR} P A_R^{-1} R Q} = \boxed{\Sigma_{LR} A_L^{-1} Q}, \quad (46)$$

$$\boxed{\Sigma'_{RL} = Q A_R^{-1} \Sigma_{RL}} = \boxed{Q R A_L^{-1} \eta_1 P \Sigma_{RL} + Q \Sigma_{RL}}, \quad (47)$$

$$\boxed{\Sigma'_{RR} = \eta_2 P + Q A_R^{-1} R Q} = \boxed{\eta_2 P + Q R A_L^{-1} Q}. \quad (48)$$

For example,

$$Q A_R^{-1} \Sigma_{RL} = Q \Sigma_{RL} - \frac{Q R P \Sigma_{RL}}{d}, \quad \eta_1 \Sigma_{LR} P A_R^{-1} \Sigma_{RL} = -\frac{\Sigma_{LR} P \Sigma_{RL}}{d},$$

so the lower-left and upper-left blocks in Eqs. (47) and (45) reduce exactly to Eqs. (34) and (32). The remaining two reductions are identical:

$$\eta_1 \Sigma_{LR} P A_R^{-1} R Q = -\frac{\Sigma_{LR} P R Q}{d}, \quad Q A_R^{-1} R Q = Q R Q - \frac{Q R P R Q}{d}.$$

Equations (45)–(48) are useful computationally because they replace the explicit scalar division by two linear solves with the same nearly identity rank-one matrix. In the right-resolvent form, solve

$$A_R X = \Sigma_{RL}, \quad A_R Y = R Q. \quad (49)$$

Then the update is assembled as

$$\begin{aligned} \Sigma'_{LL} &= \Sigma_{LL} + \eta_1 \Sigma_{LR} P X, & \Sigma'_{LR} &= \Sigma_{LR} Q + \eta_1 \Sigma_{LR} P Y, \\ \Sigma'_{RL} &= Q X, & \Sigma'_{RR} &= \eta_2 P + Q Y. \end{aligned} \quad (50)$$

The solves in Eq. (49) may be evaluated with the same Tikhonov-regularized linear solver used for the ordinary covariance updates. On well-conditioned nonsingular branches the unregularized solve gives the exact identities above, while the Tikhonov fallback supplies a controlled numerical continuation when A_R is nearly singular.

C. Off-Diagonal Choi Block

The upper-right block deserves separate emphasis:

$$\Sigma'_{LR} = \Sigma_{LR}Q - \frac{\Sigma_{LR}PRQ}{d}. \quad (51)$$

The first term removes the outgoing column in the measured P -direction: $\Sigma'_{LR}P = 0$. The second term is a finite rank-one correction created by pre-existing correlations between the selected mode and $\text{Ran } Q$. If $PRQ = 0$, meaning that the selected mode is uncorrelated with its orthogonal complement in R , then

$$\Sigma'_{LR} = \Sigma_{LR}Q. \quad (52)$$

The analogous statement for the lower-left block follows either directly or by Hermiticity.

D. Hermiticity and the Singular Branch Condition

When $R = R^\dagger$, $\Sigma_{RL} = \Sigma_{LR}^\dagger$, and $\eta_{1,2}$ are real, d is real. It then follows immediately from Eq. (31) that

$$\Sigma'_{LL} = (\Sigma'_{LL})^\dagger, \quad \Sigma'_{RR} = (\Sigma'_{RR})^\dagger, \quad \Sigma'_{RL} = (\Sigma'_{LR})^\dagger. \quad (53)$$

If $d = 0$, the rank-one denominator in the regulated contraction vanishes. The ordinary finite limit in Proposition 1 is then not defined. At the operator level this identifies a singular selected branch: the projected mode has exactly the covariance value at which the chosen local Kraus update has vanishing or nonregular overlap with the incoming accumulated tensor. Such an event requires a separate zero-overlap or rescaled-limit analysis and is not covered by the formula above.

E. Preservation of the Choi Involution

For a pure Gaussian Choi tensor, the covariance is an involution,

$$\Sigma^2 = \mathbb{I}. \quad (54)$$

The corrected regulated update preserves this constraint on every nonsingular branch.

Proposition 2 (Preservation of the Choi involution). *Assume the hypotheses of Proposition 1, together with Eq. (54). Then, for $d = \text{Tr}(P\Sigma_{RR}) - \eta_1 \neq 0$,*

$$(\Sigma')^2 = \mathbb{I}. \quad (55)$$

Proof. The assertion is basis independent, so for the proof choose a basis adapted to $P \oplus Q$, and reorder the full covariance as $(L \oplus Q) \oplus P$. Writing

$$\Sigma_{LR} = \begin{pmatrix} X & y \\ u & V \end{pmatrix}, \quad \Sigma_{RR} = \begin{pmatrix} r & b^\dagger \\ b & C \end{pmatrix}, \quad r = \text{Tr}(P\Sigma_{RR}), \quad (56)$$

the unupdated covariance becomes

$$\tilde{\Sigma} = \begin{pmatrix} X & y \\ y^\dagger & r \end{pmatrix}, \quad X = \begin{pmatrix} \Sigma_{LL} & V \\ V^\dagger & C \end{pmatrix}, \quad y = \begin{pmatrix} u \\ b \end{pmatrix}. \quad (57)$$

Equation (54) is equivalent to

$$X^2 + yy^\dagger = \mathbb{I}_{L \oplus Q}, \quad Xy + ry = 0, \quad y^\dagger y + r^2 = 1. \quad (58)$$

Since $\tilde{\Sigma}$ is Hermitian, the second identity also gives $y^\dagger X = -ry^\dagger$.

The corrected update in Eq. (31), written in the same reordered output basis, decouples the newly reset P mode:

$$\tilde{\Sigma}' = \begin{pmatrix} T & 0 \\ 0 & \eta_2 \end{pmatrix}, \quad T = X - \frac{yy^\dagger}{d}, \quad d = r - \eta_1. \quad (59)$$

Using Eq. (58),

$$\begin{aligned} T^2 &= X^2 - \frac{Xyy^\dagger + yy^\dagger X}{d} + \frac{(yy^\dagger)^2}{d^2} \\ &= \mathbb{I}_{L \oplus Q} + \left(-1 + \frac{2r}{d} + \frac{1-r^2}{d^2} \right) yy^\dagger \\ &= \mathbb{I}_{L \oplus Q}, \end{aligned} \quad (60)$$

because

$$-d^2 + 2rd + 1 - r^2 = 1 - \eta_1^2 = 0. \quad (61)$$

Finally, $\eta_2^2 = 1$, and therefore $(\tilde{\Sigma}')^2 = \mathbb{I}$. Permuting back to the original Choi-leg ordering proves Eq. (55). $\square$

Thus an admissible local branch maps a pure Choi covariance to another pure Choi covariance: the selected mode is reset to an exact η_2 eigenmode, while the complementary block remains an involution through the finite Schur-complement update.

VII. PARTICLE-HOLE TRANSFER COMPLEMENTARITY OF A PURE CHOI COVARIANCE

The accumulated covariance Σ is itself the covariance of a pure Gaussian Choi state. It therefore contains a single-particle transfer description of the realized non-unitary operator. The natural language is not that of a failure of invertibility under projection, but that of complementary particle and hole amplitudes: an exact projected direction can be dark for particle propagation while it is a pole of the hole representation, or vice versa.

In this section the two Choi legs are equal-dimensional one-particle spaces,

$$\dim \mathcal{H}_A = \dim \mathcal{H}_B = N, \quad (62)$$

and the symbols A, B, D denote Choi-covariance blocks. They are distinct from the rank-one measurement projector P of Eqs. (4)–(31).

A. Pure covariance identities

Let $|\Sigma\rangle$ denote the pure number-conserving Choi state, with

$$C_{ij} = \langle \Sigma | c_i^\dagger c_j | \Sigma \rangle, \quad \Sigma_{ij} = \langle \Sigma | [c_i^\dagger, c_j] | \Sigma \rangle = 2C_{ij} - \delta_{ij}. \quad (63)$$

The purity condition $C^2 = C$ is equivalent to

$$\Sigma = \Sigma^\dagger, \quad \Sigma^2 = \mathbb{I}. \quad (64)$$

After acting on the Choi state, creation and annihilation operators obey

$$c^\dagger |\Sigma\rangle = -\Sigma c^\dagger |\Sigma\rangle, \quad c |\Sigma\rangle = \Sigma^T c |\Sigma\rangle. \quad (65)$$

These are state identities and are not operator equalities on the full Fock space.

B. Consistency of the Choi-state identities

We now verify explicitly that Eq. (65) encodes the same complex covariance Σ used in Eq. (63). Sandwiching the annihilation identity between $\langle \Sigma | c_i^\dagger$ and $|\Sigma\rangle$ gives

$$\begin{aligned} C_{ij} &= \langle \Sigma | c_i^\dagger c_j | \Sigma \rangle = \sum_k \Sigma_{kj} \langle \Sigma | c_i^\dagger c_k | \Sigma \rangle \\ &= \sum_k C_{ik} \Sigma_{kj}, \quad \implies \quad C = C\Sigma. \end{aligned} \quad (66)$$

Because both C and Σ are Hermitian, taking the adjoint yields

$$\Sigma C = C. \quad (67)$$

Thus the occupied one-particle subspace lies in the $+1$ eigenspace of Σ .

Similarly, use the creation identity and contract with $\langle \Sigma | c_i$. Since $\langle \Sigma | c_i c_j^\dagger | \Sigma \rangle = \delta_{ij} - C_{ji}$, one obtains

$$\delta_{ij} - C_{ji} = - \sum_k \Sigma_{jk} (\delta_{ik} - C_{ki}). \quad (68)$$

Upon transposing this matrix equation,

$$\Sigma(\mathbb{I} - C) = -(\mathbb{I} - C). \quad (69)$$

Hence the empty one-particle subspace lies in the -1 eigenspace of Σ . Resolving the identity into occupied and empty sectors now gives

$$\Sigma = \Sigma C + \Sigma(\mathbb{I} - C) = C - (\mathbb{I} - C) = 2C - \mathbb{I}. \quad (70)$$

Therefore

$$\boxed{\langle \Sigma | [c_i^\dagger, c_j] | \Sigma \rangle = 2\langle \Sigma | c_i^\dagger c_j | \Sigma \rangle - \delta_{ij} = \Sigma_{ij}.} \quad (71)$$

This closes the consistency loop: the linear relations in Eq. (65) are precisely the state-space representation of the complex Choi covariance, rather than an additional transfer assumption.

In the bipartition $\mathcal{H}_A \oplus \mathcal{H}_B$, write

$$\Sigma = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A = A^\dagger, \quad D = D^\dagger. \quad (72)$$

Equation (64) gives

$$\begin{aligned} A^2 + BB^\dagger &= \mathbb{I}_A, & AB + BD &= 0, \\ B^\dagger A + DB^\dagger &= 0, & B^\dagger B + D^2 &= \mathbb{I}_B. \end{aligned} \quad (73)$$

In particular,

$$\mathbb{I}_A - A^2 = BB^\dagger \geq 0. \quad (74)$$

Since A is Hermitian, it admits an orthonormal eigenbasis and

$$\text{spec}(A) \subseteq [-1, 1]. \quad (75)$$

C. Particle and hole transfer matrices

Taking the A -leg components of Eq. (65) yields

$$(\mathbb{I}_A + A)\mathbf{c}_A^\dagger |\Sigma\rangle = -B\mathbf{c}_B^\dagger |\Sigma\rangle, \quad (76)$$

$$(\mathbb{I}_A - A^T)\mathbf{c}_A |\Sigma\rangle = B^*\mathbf{c}_B |\Sigma\rangle. \quad (77)$$

Accordingly, wherever the indicated finite representation exists, define

$$\mathbf{c}_A^\dagger|\Sigma\rangle = \mathcal{T}_p \mathbf{c}_B^\dagger|\Sigma\rangle, \quad \mathcal{T}_p := -(\mathbb{I}_A + A)^{-1}B, \quad (78)$$

$$\mathbf{c}_A|\Sigma\rangle = \mathcal{T}_h \mathbf{c}_B|\Sigma\rangle, \quad \mathcal{T}_h := (\mathbb{I}_A - A^T)^{-1}B^*. \quad (79)$$

Here $\mathcal{T}_p$ is the particle, or creation-sector, transfer matrix, while $\mathcal{T}_h$ is the hole, or annihilation-sector, transfer matrix. Neither definition requires A itself to have full rank, and a rank-deficient $\mathcal{T}_p$ is entirely regular. Rather, $\mathcal{T}_p$ develops a pole only at an exact $A = -\mathbb{I}$ direction, whereas $\mathcal{T}_h$ develops the complementary pole at an exact $A = +\mathbb{I}$ direction. At such endpoints the displayed inverse formulas are understood by approach from the finite sector.

Proposition 3 (Particle-transfer representation). *On a finite particle-transfer sector, Eq. (78) implies*

$$A = (\mathbb{I}_A - \mathcal{T}_p \mathcal{T}_p^\dagger)(\mathbb{I}_A + \mathcal{T}_p \mathcal{T}_p^\dagger)^{-1}, \quad (80)$$

$$B = -2(\mathbb{I}_A + \mathcal{T}_p \mathcal{T}_p^\dagger)^{-1} \mathcal{T}_p. \quad (81)$$

On the corresponding coupled subspace of $\mathcal{H}_B$, one furthermore has

$$D = -(\mathbb{I}_B - \mathcal{T}_p^\dagger \mathcal{T}_p)(\mathbb{I}_B + \mathcal{T}_p^\dagger \mathcal{T}_p)^{-1}. \quad (82)$$

Proof. Equation (78) gives

$$B = -(\mathbb{I}_A + A)\mathcal{T}_p. \quad (83)$$

Because $A = A^\dagger$, substitution into $A^2 + BB^\dagger = \mathbb{I}_A$ first gives the unsimplified relation

$$(\mathbb{I}_A + A)\mathcal{T}_p \mathcal{T}_p^\dagger (\mathbb{I}_A + A) = \mathbb{I}_A - A^2. \quad (84)$$

On a finite particle-transfer sector, $\mathbb{I}_A + A$ is invertible. Multiplying Eq. (84) on both sides by $(\mathbb{I}_A + A)^{-1}$ therefore produces the two-inverse expression

$$\mathcal{T}_p \mathcal{T}_p^\dagger = (\mathbb{I}_A + A)^{-1}(\mathbb{I}_A - A^2)(\mathbb{I}_A + A)^{-1}. \quad (85)$$

Now A , $\mathbb{I}_A - A$, $\mathbb{I}_A + A$, and $(\mathbb{I}_A + A)^{-1}$ mutually commute, because each is a function of the Hermitian operator A . Factoring $\mathbb{I}_A - A^2 = (\mathbb{I}_A - A)(\mathbb{I}_A + A)$ and cancelling one factor yields

$$\mathcal{T}_p \mathcal{T}_p^\dagger = (\mathbb{I}_A - A)(\mathbb{I}_A + A)^{-1}. \quad (86)$$

The cancellation is unavailable precisely on an $a = -1$ eigenvector, where $\mathbb{I}_A + A$ vanishes and the particle-transfer representation has its endpoint pole. Inverting this Cayley relation yields Eq. (80); inserting the result into Eq. (83) yields Eq. (81). Finally, $AB + BD = 0$ fixes D on $\text{Ran } B^\dagger$, giving Eq. (82). $\square$

It is important that the qualifier in the last statement not be suppressed. If an exact projection makes B vanish on a direction, the equation $AB + BD = 0$ cannot determine the polarization of the decoupled mode on the opposite Choi leg: there one has only $D^2 = \mathbb{I}_B$. Thus Eqs. (80) and (81) determine particle transfer data, whereas a fully projected endpoint retains additional output-polarization information in D . This is precisely the information carried by the $\eta_2 P$ reset term in Eq. (31).

D. Polar form and particle-hole duality

Take the left polar decomposition

$$\mathcal{T}_p = HU, \quad H = (\mathcal{T}_p \mathcal{T}_p^\dagger)^{1/2} \geq 0. \quad (87)$$

Because the Choi legs have equal dimension, the partial isometry can be extended to a unitary $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$. On the particle-transfer sector,

$$A = q(H) := \frac{\mathbb{I} - H^2}{\mathbb{I} + H^2}, \quad B = -r(H)U, \quad r(H) := \frac{2H}{\mathbb{I} + H^2}. \quad (88)$$

The functions $q(H)$ and $r(H)$ commute and obey

$$q(H)^2 + r(H)^2 = \mathbb{I}_A, \quad (89)$$

making purity transparent on the coupled sector.

For strictly positive finite singular values of H , substitution into Eq. (79) gives

$$\mathcal{T}_h = -(H^T)^{-1} U^* = -H^{-T} U^*. \quad (90)$$

Thus particle and hole amplitudes are dual descriptions of the same pure Choi correlation:

$$s_j(\mathcal{T}_h) = s_j(\mathcal{T}_p)^{-1} \quad \text{for } 0 < s_j(\mathcal{T}_p) < \infty. \quad (91)$$

This is particle-hole complementarity in transfer space, rather than an alternative convention appended to the dynamics.

E. Endpoint sectors and Lyapunov exponents

Let

$$A u_j = a_j u_j, \quad a_j \in [-1, 1]. \quad (92)$$

Equation (86) gives, on the finite sector,

$$s_{p,j}^2 = \frac{1 - a_j}{1 + a_j}, \quad s_{h,j}^2 = \frac{1 + a_j}{1 - a_j}. \quad (93)$$

Consequently the endpoint structure is

$$a_j = +1 : \quad s_{p,j} = 0, \quad s_{h,j} = \infty, \quad (94)$$

$$-1 < a_j < +1 : \quad 0 < s_{p,j} < \infty, \quad s_{h,j} = s_{p,j}^{-1}, \quad (95)$$

$$a_j = -1 : \quad s_{p,j} = \infty, \quad s_{h,j} = 0. \quad (96)$$

The infinite values in Eqs. (94) and (96) are limiting statements: at the exact endpoint the relevant inverse in Eq. (78) or (79) is singular. Exact projective operations polarize selected one-particle directions into these extremal covariance sectors. A particle-dark direction is a pole in the complementary hole description, and a hole-dark direction is a pole in the particle description. This is not a loss of purity or a failure of the Choi covariance; it is the sharp transfer signature of an exact Gaussian projection.

For an accumulated Choi tensor after $t > 0$ completed cycles, the finite particle and hole Lyapunov exponents are therefore

$$\lambda_{p,j}(t) = \frac{1}{t} \log s_{p,j}(t) = \frac{1}{2t} \log \left(\frac{1 - a_j(t)}{1 + a_j(t)} \right) = -\frac{1}{t} \operatorname{artanh} a_j(t), \quad (97)$$

$$\lambda_{h,j}(t) = \frac{1}{t} \log s_{h,j}(t) = -\lambda_{p,j}(t). \quad (98)$$

The endpoint sectors contribute formal $-\infty$ or $+\infty$ exponents in the appropriate particle or hole representation. A supported numerical observable may omit these endpoint sectors and record their multiplicities separately, but it must identify the transfer convention to which its retained exponents refer.

F. Particle-transfer Lyapunov gap

The transfer-space gap relevant for the slowest finite particle direction is the finite-time Lyapunov gap

$$\Delta_\lambda(t) \equiv \min_{j \text{ finite}} |\lambda_{p,j}(t)| = \frac{1}{t} \min_{j \text{ finite}} |\log s_j(\mathcal{T}_p(t))|. \quad (99)$$

By Eq. (97), it is obtained directly from the left Choi block:

$$\Delta_\lambda(t) = \frac{1}{t} \min_{j \text{ finite}} |\operatorname{artanh} a_j(t)|, \quad a_j(t) \in \operatorname{spec} \Sigma_{LL}(t). \quad (100)$$

On the open interval $(-1, 1)$, the function $|\operatorname{artanh} a|$ is strictly increasing with $|a|$. Thus the finite gap is controlled by eigenvalues of $A(t) = \Sigma_{LL}(t)$ closest to zero, rather than by its polarized endpoint sectors. The latter have $a_j = \pm 1$ and contribute formal infinite particle exponents; they are excluded from the finite minimum in Eq. (100) while their multiplicities remain a separate diagnostic of exact projection.

Equivalently, if $Au_j = a_j u_j$, then Eq. (86) gives

$$\mathcal{T}_p \mathcal{T}_p^\dagger u_j = \frac{1 - a_j}{1 + a_j} u_j = s_{p,j}^2 u_j. \quad (101)$$

The A -eigenstates closest to $a_j = 0$ are therefore precisely the eigenstates of $\mathcal{T}_p \mathcal{T}_p^\dagger$ whose eigenvalues are closest to unity. These are the near-gap transfer states.

It is important not to replace this criterion by $\sigma_{\min}(\mathbb{I} - \mathcal{T}_p)$. The latter probes whether the operator $\mathcal{T}_p$ has an approximately fixed vector, whereas Eq. (99) probes whether a singular value of $\mathcal{T}_p$ lies near unity. For a non-normal transfer operator these questions are inequivalent. Their identification would require additional normality and alignment assumptions that do not follow from purity of the Choi covariance.

G. Information content of the Choi blocks

For the particle Lyapunov spectrum, the upper-left Choi block is already complete data. Indeed, Eq. (86) states that

$$\mathcal{T}_p \mathcal{T}_p^\dagger = (\mathbb{I}_A - A)(\mathbb{I}_A + A)^{-1}. \quad (102)$$

Consequently the eigenvalues of $A = \Sigma_{LL}$ determine all particle singular values, while its eigenvectors are precisely the eigenstates of the positive transfer operator $\mathcal{T}_p \mathcal{T}_p^\dagger$ (equivalently, the left singular vectors of $\mathcal{T}_p$). This includes the regular zero-eigenvalue sector at $a = +1$. Thus neither an explicit construction of $\mathcal{T}_p$ nor a spectral decomposition of $B = \Sigma_{LR}$ is required in order to compute $\lambda_{p,j}(t)$ and the corresponding $\mathcal{T}_p \mathcal{T}_p^\dagger$ eigenstates.

For a complete particle spectrum one may diagonalize $A = \Sigma_{LL}$. For the gap alone, however, a complete intermediate spectral resolution is unnecessary. Since the near-gap states minimize $|a_j|$, they are the lowest-eigenvalue states of the positive operator

$$A^2 = \Sigma_{LL}^2. \quad (103)$$

An iterative Hermitian eigensolver can access this sector using only the matrix-vector operation

$$v \mapsto A(Av), \quad (104)$$

without materializing A^2 , $\mathcal{T}_p$, or $\mathcal{T}_p \mathcal{T}_p^\dagger$. Because A^2 does not distinguish the signs of paired eigenvalues $+a$ and $-a$, the signed particle exponents must be recovered by diagonalizing the projected Hermitian operator $X^\dagger A X$ inside the converged near-zero subspace X . An exact diagonalization of A at the final observation time then provides the complete final spectrum and an independent validation of the iterative near-gap extraction.

The off-diagonal block has different, equally physical information content. In the polar representation of Eq. (88), A fixes the positive factor H , whereas

$$B = -r(H)U \quad (105)$$

fixes the partial-isometry, or angular, data U on the coupled support. Therefore B is required to reconstruct the full complex transfer matrix, its input-leg spectral data, or the mapping between input-leg and output-leg orbitals. The distinction is operationally useful: a calculation whose target is only the Lyapunov spectrum and the output-leg eigenstates of $\mathcal{T}_p \mathcal{T}_p^\dagger$ may diagonalize Σ_{LL} alone, whereas an analysis of the transported wavefunctions themselves must retain Σ_{LR} .

H. Regularity of a zero particle-transfer sector

Rank deficiency of $\mathcal{T}_p$ itself is regular. Let Π_0 be the orthogonal projector onto $\ker H$. Then

$$q(H)\Pi_0 = \Pi_0, \quad r(H)\Pi_0 = 0, \quad (106)$$

so $H = 0$ is precisely the $a = +1$, particle-dark endpoint. If desired, a full-rank family approaching this endpoint may be introduced by

$$H_\varepsilon = H + \varepsilon\Pi_0, \quad \varepsilon > 0. \quad (107)$$

Restricted to $\text{Ran } \Pi_0$,

$$q(H_\varepsilon)\Pi_0 = (1 - 2\varepsilon^2 + O(\varepsilon^4))\Pi_0, \quad (108)$$

$$r(H_\varepsilon)\Pi_0 = (2\varepsilon + O(\varepsilon^3))\Pi_0. \quad (109)$$

The particle amplitude therefore vanishes continuously while its hole partner diverges in accordance with Eq. (91).

Remark 1. The matrices $\mathcal{T}_p$ and $\mathcal{T}_h$ above are complex, number-conserving particle and hole transfer matrices extracted from Σ . A direct complex-basis GPU observable associated with `colab_regularized_choi_transfer_matrix/src/choi_transfer_observables_gpu.py` uses the particle-spectrum consequence of Eq. (102) in the complex basis. For time-resolved gap extraction it may target the near-zero spectrum of A^2 through Eq. (104), while an exact final-time diagonalization of A supplies the full final particle spectrum and selected near-gap eigenstates of $\mathcal{T}_p\mathcal{T}_p^\dagger$. No Majorana covariance, pseudoinverse, or explicit stored $\mathcal{T}_p$ is required. The complementary finite hole exponents are recovered by Eq. (98).

VIII. RELATION TO THE REPOSITORY DYNAMICS

The cumulative-Choi construction developed here is useful when one intends to retain a composed non-unitary operator and, for example, extract a transfer spectrum from its Choi blocks. Its complex blocks encode the complementary particle and hole transfer data described in Sec. VII. The dedicated analysis bundle, `colab_regularized_choi_transfer_matrix/`, propagates the corrected cumulative Choi blocks. The intended finite-size gap campaign uses

$$N_x = 20, \quad N_y \in \{30, 40, 50\}, \quad T = 2N_y, \quad (110)$$

and tracks

$$\Delta_\lambda(t) = \min_j |\lambda_{p,j}(t)| \quad (111)$$

at each completed cycle. In the domain-wall truncated geometry the observable is evaluated on the active slab-only Choi basis; in the homogeneous $DW = \text{false}$ geometry it is evaluated on the complete top-layer Choi basis. The numerical strategy is to extract the near-gap sector through the matrix-free A^2 procedure above throughout the trajectory, and at $t = T$ to retain an exact particle spectrum and selected near-gap eigenstates of $\mathcal{T}_p\mathcal{T}_p^\dagger$. This describes the intended direct complex-basis observable and does not assert that these finite-size campaign products have already been generated.

The canonical Markov simulations in this repository take a different route. The CPU and GPU implementations of `classA_U1FGTN.run_markov_circuit(...)` and `classA_U1FGTN_gpu.run_markov_circuit(...)` propagate the normalized top-layer covariance G_t , not Σ_t . For finite Wannier support, their rank-one measurement maps are evaluated by solving on the compact support block of the local mode. Those support-reduced formulas are derived in `src/fgtn/support_reduced_markov_update_note.tex`. The stochastic covariance process and its tangent cocycle are organized in `markov_transfer_operators/markov_transfer_operators_note.tex`.

Accordingly, distinct constructions should not be conflated:

1. The present ε -prescription is an analytic definition of a singular cumulative-Choi contraction before taking $\varepsilon \rightarrow 0$.
2. Endpoint poles of $\mathcal{T}_p$ or $\mathcal{T}_h$ express particle-hole complementarity under an exact projective update.
3. The direct complex particle-spectrum observable extracts only $\mathcal{T}_p\mathcal{T}_p^\dagger$ data; reconstructing the full transfer matrix or its input-leg spectral data requires the additional angular information in Σ_{LR} .

Only the first is required to define the regulated contraction itself.

IX. NUMERICAL CONSISTENCY CHECK

The basis-independent result can be verified without using any circuit simulation data. For each test one generates Hermitian Σ_{LL} and $R = \Sigma_{RR}$, an arbitrary Σ_{LR} with $\Sigma_{RL} = \Sigma_{LR}^\dagger$, a rank-one projector P , and each of the four sign pairs (η_1, η_2) . Cases with d close to zero are excluded. The directly regulated contraction obtained at finite ε from Eqs. (6) and (7) is then compared to Eq. (31).

For representative random matrices with $N = 3, 5, 7$, the largest Frobenius discrepancy over the four sign pairs was Hermiticity of the output blocks is also satisfied to floating-point precision, in agreement with Eq. (53).

TABLE II. Maximum full-Choi-block discrepancy between the regulated finite- ε contraction and the basis-independent $\varepsilon \rightarrow 0$ formula for a fixed random test ensemble. The linear decrease with ε is consistent with the first omitted term in the regulated expansion.

Dimension N	$\varepsilon = 10^{-4}$	$\varepsilon = 10^{-6}$	$\varepsilon = 10^{-8}$
3	1.9×10^{-3}	1.9×10^{-5}	1.9×10^{-7}
5	9.0×10^{-3}	9.0×10^{-5}	9.0×10^{-7}
7	2.6×10^{-2}	2.6×10^{-4}	2.6×10^{-6}

The involution statement was tested separately by generating random Hermitian involutions $\Sigma = U \text{diag}(\pm 1) U^\dagger$, random rank-one projectors P , and applying all four sign branches while excluding $|d| < 10^{-8}$. For $N = 3, 5, 8$, with 100 samples per dimension and 1200 nonsingular branch updates in total, the worst residuals were

$$\max \left\| (\Sigma')^2 - \mathbb{I} \right\|_F = 7.317 \times 10^{-15}, \quad \max \left\| \Sigma' - (\Sigma')^\dagger \right\|_F = 8.689 \times 10^{-16}. \quad (112)$$

These errors are at floating-point roundoff and provide an independent check of the algebraic preservation proof.

X. CONCLUSION

The local adaptive Kraus branches of the class-A fermionic circuit admit a compact two-sign Choi representation, but their rank-one projector structure makes naive full-tensor composition singular. The regulated contraction $K_{LL}^{(\varepsilon)} = \eta_1 P + \varepsilon Q$ supplies a controlled limit and exposes a simple basis-independent answer. The selected mode is reset by $\eta_2 P$; all other modes are updated by an embedded Schur complement; and the inter-leg Choi correlations receive a finite correction determined by the correlation of the selected mode with its complement. Crucially, the lower-right reduced Schur complement must be embedded into $\text{Ran } Q$ before it is combined with the full-space P -sector contribution. This resolves the dimensional ambiguity in the handwritten expression and gives a complete operator-valued update suitable for cumulative Choi transfer analyses. When the input covariance is pure, $\Sigma^2 = \mathbb{I}$, the corrected update also preserves the involution exactly on every nonsingular branch. Finally, the bipartite Choi covariance resolves the transfer content into conjugate particle and hole descriptions. A rank-deficient particle transfer is a regular, particle-dark sector; its complementary hole representation has an endpoint pole, and the opposite statement holds at the other extremal covariance polarization. Any supported numerical spectrum must therefore state its basis and whether it retains particle, hole, or Majorana transfer data.